MIL-P-2610B
28 SEPTEMBER 1964
SUPERSEDING
MIL-P-2610A
29 APRIL 1966

MILITARY SPECIFICATION

POLISH, METAL, (LIQUID: MIRROR, SEARCHLIGHT: NONFREEZING TYPE)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade and type polish for cleaning rhodium plated and other metal mirrors (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-E-751 — Ether, Petroleum; Technical-Grade.

RR-S-366 — Sieves, Standard for Testing Purposes.

PPP-B-636 — Box, Fiberboard.

PPP-C-186 — Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

MILITARY

JAN-C-293 — Calcium Carbonate (For Use in Other Than Paint Products).

MIL-F-5566 — Fluid; Anti-Icing (Isopropyl Alcohol.)

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7930

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, one filled primary container of the finished item shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.1.1. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.2 Material. Material shall be as specified herein and shall be free from impurities which will decrease the reflectivity of the metal surface.

*** 3.3 Composition.** The composition of the metal polish exclusive of the suspending agent (see 3.4) shall be as shown in table I.

TABLE I. *Composition of metal polish.*

Component	Percent by weight
Calcium carbonate	7.5 ± 1.0
Rouge	0.27 ± 0.04
Black nigrosine (water soluble)	0.004 ± 0.0005
Isopropyl alcohol	74.5 ± 1.5
Water	13.4 ± 1.0
Petroleum ether	4.3 ± 0.1

3.3.1 *Component materials.* Calcium carbonate shall conform to JAN–C–293; isopropyl alcohol shall conform to MIL–F–5566; petroleum ether shall conform to O–E–751.

3.3.2 *Rouge.* The rouge components shall be as specified in table II.

TABLE II. *Range components.*

	Percent by weight	
	Maximum	Minimum
Ferric oxide		94.8
Silicon dioxide	5.0	
Grit	0.20	

*** 3.3.3** *Total grit content of polish.* The grit content shall not exceed 0.007 percent of the total polish weight when tested as specified in 4.3.2.

3.4 Suspending agent. A suspending agent may be added to the product (see 6.3. Not more than 3 percent of suspending agent based on total polish weight shall be added. Only those suspending agents which contain a minimum of 80 percent silicon dioxide shall be used (see 6.3.1). The product containing the suspending agent shall meet all requirements of this specification.

*** 3.5 Fineness of polish solids.** Not more than 0.5 percent of the solids shall be retained on a U. S. Standard No. 325 sieve when tested as specified in 4.3.2.

*** 3.6 Polish solids.** The solids portion of the metal polish shall comprise no less than 7.5 percent of the total polish weight when tested as specified in 4.3.2.

*** 3.7 Performance.** The metal polish shall produce a minimum of a 16 gloss unit increase in the reflectivity of a rhodium plated surface when compared to the standard cleaner, shall not scratch, and shall leave no residue when tested as specified in 4.3.2.

*** 3.8 Marking.** Each unit container of the metal polish shall be labelled or stencilled with the following information:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

CAUTION

HIGHLY FLAMMABLE LIQUID (CONTAINS PETROLEUM ETHER AND ISOPROPYL ALCOHOL).

STORE IN A COOL AREA.

DO NOT USE NEAR AN OPEN FLAME.

KEEP TIGHTLY CAPPED.

DO NOT TAKE INTERNALLY.

USE WITH ADEQUATE VENTILATION.

INSTRUCTIONS FOR USE

SHAKE WELL BEFORE USING.

LIGHTLY BRUSH ALL LOOSE DIRT FROM MIRROR WITH A CLEAN SOFT CLOTH.

APPLY POLISH SPARINGLY TO A CLEAN SOFT CLOTH.

RUB LIGHTLY OVER A SMALL (1/8 OR LESS) SECTION OF THE MIRROR SURFACE.

CLEAN AND POLISH WITH A CLEAN SOFT CLOTH WHILE POLISH IS DAMP. DO NOT ALLOW POLISH TO DRY ON MIRROR.

REPEAT UNTIL ENTIRE MIRROR IS CLEAN.

* **3.9 Workmanship.** The polish shall be free of foreign particles. The finished product shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable quality levels in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Preacceptance inspection.* Inspection of the preproduction sample shall be conducted to determine compliance with all the requirements of this specification.

4.2 Sampling for inspection. Sampling for inspection shall be in accordance with the provisions of MIL–STD–105, except where otherwise indicated.

4.2.1 *Inspection of components and materials.* The supplier shall be responsible for insuring that the quality assurance provisions for component materials shall be in accordance with this specification and with referenced subsidiary specifications to the extent applicable. A supplier's certificate of compliance will be acceptable for the following:

a. The rouge material and composition requirements specified in 3.3.2.

b. The requirements for calcium carbonate, isopropyl alcohol and petroleum ether specified in 3.3.1.

c. Suspending agent composition if used (see 3.4).

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.1.5. Random samples shall be drawn from each lot of end items in the same size container for examination for visual contents, and preparation for delivery defects. The lot size for purposes of determining the sample size, in accordance with MIL–STD–105, shall be expressed in units of filled unit containers of the same capacity, consisting of polish manufactured

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

in one batch for examination in 4.3.1.1, 4.3.1.2 and 4.3.1.3, and in units of shipping containers for examination under 4.3.1.4.

4.3.1.1 *Examination of the unit container.* The sample unit for this examination shall be one filled unit container.

Examine	*Defect*
Construction	Not type specified (see 5.1.1.1).
Closure and seal	Not type specified (see 5.1.1.1).
Workmanship	Broken, chipped, dirty; any other defect affecting serviceability; any leaking of contents.
Markings	Missing, incomplete, incorrect, not as specified (see 3.8).

4.3.1.2 *Examination of contents of the unit container.* The sample unit for this examination shall be the contents of one filled unit container. The sample shall be transferred into a clear glass container and after standing for three hours shall be examined for the presence of foreign particles visible to the naked eye.

Examine	*Defect*
Workmanship	Contains foreign particles.

4.3.1.3 *Examination of the end item for net contents.* The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container is less than specified or indicated for all sample units examined.

4.3.1.4 *Examination of the end item for defects in preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship and contents comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers, fully prepared for delivery, shall be examined for closure defects.

Examine	*Defect*
Markings	Incorrect, incomplete, illegible, omitted; of improper size, sequence or method of application; not in accordance with contract requirements.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling.
	Bulged or distorted containers.
Contents	Number of intermediate packages per shipping container is more or less than required.
	Number of unit containers per intermediate package is less than specified.

4.3.1.5 *Inspection levels and acceptable quality levels (AQL's) for examination.* The inspection levels and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

Examination paragraph	*Inspection level*	*AQL*
4.3.1.1	I	2.5
4.3.1.2	S–4	1.0
4.3.1.3	S–2	....
4.3.1.4	S–2	2.5

* 4.3.2 *Testing of the end item.* The end item shall be tested for the characteristics specified in Section 3 in accordance with test methods specified in table III. The lot size for test purposes shall be expressed in gallons of polish. The sample unit shall be in two pints of polish. The sample size for tests shall be as indicated below. The sample units shall be randomly selected. The lot shall be unacceptable if there is any failure of any sample unit to meet any characteristic specified.

Lot size (unit containers)	*Sample size*
800 or less	2
801 to 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

TABLE III. *Instructions for testing of the end item.*

Characteristic	Specification reference		Requirements applicable to		Number determinations per/unit sample	Results reported as		Inspect level	AQL
	Requirement	Test method	Individ unit	Lot aver		Pass or fail	Numerically to nearest		
Composition	3.3	Note [1]							
Total grit content of polish, percent.	3.3.3	4.4.3	X		1		0.001		
Suspending agent	3.4	Note [1]							
Fineness of polish solids, percent.	3.5	4.4.2	X		1		0.1		
Polish solids, percent.	3.6	4.4.1	X		1		0.1		
Performance	3.7	4.4.4	X		1		1		

Note: [1] The supplier shall submit a certificate of compliance for this characteristic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

4.4 Test procedures.

* 4.4.1 *Solias content.* Weigh to the nearest 0.1 gram, one-half of the contents from a bottle of polish which has just been thoroughly mixed. Separate the solid material by filtering the sample through a tared No. 41 Whatman or equivalent filter paper. Wash the solids with acetone and air dry for 15 minutes. Dry at 105° ± 2°C. to a constant weight (nearest 0.02 grams)

4.4.1.1 *Calculation.* $C = \frac{A}{B} \times 100$

Where A = Weight of dry solids minus weight of filter paper.

B = Weight of polish sample.

C = Percent of total solids in sample.

4.4.2 *Fineness.*

4.4.2.1 *Apparatus.*

4.4.2.1.1 *Sieves.* A nest of two sieves, the lower being a No. 325 (44-micron) and the upper a No. 16 (1190-micron) both conforming to type I, class 1 of RR–S–366.

4.4.2.1.2 *Container.* A pan or vessel of a ize sufficient to contain the sample covered with water and to permit vigorous agitation without inadvertent loss of any part of the sample or water.

4.4.2.2 *Test sample.* The polish solids obtained in 4.4.1 shall be used for this test.

4.4.2.3 *Procedure.* The test sample shall be quantitatively transferred to the container and sufficient water added to cover it. The contents of the container shall be agitated and the mixture of water and solids poured immediately over the nested sieves, arranged with the coarser sieve on top. Wash the material remaining on the sieves until the washings show no particles. The material retained on the nested sieves shall be washed into a tared beaker dried to constant weight at a temperature of 105° ± 2°C and weighed to the nearest 0.02 grams.

4.4.2.4 *Calculation.* $A = \frac{B - C}{B} \times 100$

Where A = Percent of material retained on sieves.

B = Dry weight of test sample (see 4.4.1.1).

C = Weight.

* 4.4.3 *Grit.*

4.4.3.1 *Apparatus.* The test screen shall consist of a U.S. Standard No. 200 sieve conforming to type I, class 1 of RR–S–366.

4.4.3.2 *Procedure.* The grit content shall be determined as follows: Thoroughly disperse 10.0 grams of polish solids obtained as outlined in 4.4.1 in distilled water in a 250 ml. beaker. Wash the liquid through the U.S. Standard No. 200 sieve. Washing may be assisted by the use of a fine jet of distilled water. Any rouge lumps may be broken by pressing them against the sides of the sieve. When the screening is complete, the retained material shall be washed back into the original beaker and filtered through a rapid filter paper. The paper containing the residue shall be ignited in a fused quartz crucible. The crucible shall be cooled by immersing in water. From 2 to 3 milliliters of diluted sulfuric acid shall be added and heated to fuming. The mixture shall be cooled and diluted with water, and the contents of the crucible shall be washed into the original beaker. From 5 to 10 milliliters of concentrated hydrochloric acid shall be added and digested until the ferric oxide is dissolved completely. The mixture shall be diluted with water and decanted retaining the solid residue in the beaker. The process shall be continued until the methyl-red acidity test shows the solution to be no longer acid. When the solution is free of

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

acid, the residue shall be washed on the 200 mesh screen. The residue shall then be dried and weighed, and the percent residue shall be calculated as grit.

4.4.3.2.1 *Calculation.* $D = \frac{A}{B}$ (C)

Where A = Weight of residue (grit).

B = Sample weight (polish solids).

C = Percent of total solids (C of 4.4.1.1).

D = Percent grit.

* 4.4.4 *Performance test.*

4.4.4.1 *Test materials.*

4.4.4.1.1 *Mirror.* The mirror used in this test shall be made by plating rhodium onto brass stock and polishing to a mirror bright finish (minimum of 600 gloss units). The minimum mirror size shall be 3 by 2 inches. Mirrors which contain any scratches shall not be used.

4.4.4.1.2 *Glossmeter.* The glossmeter (Photovolt Glossmeter Model 660, see 6.4.1) or equivalent shall be capable of measuring the 45 degree specular gloss of metallic surfaces in the range 0 to 800 gloss units (in terms of a perfect mirror of 1000 gloss units).

4.4.4.1.3 *Polishing apparatus.* The polishing apparatus shall consist essentially of an electric motor mounted on a flat metal plate, and a mechanism through which the motor imparts a reciprocating motion to a wool cloth polishing head lengthwise across a test panel positioned on a platform which is situated above the mechanism (see 6.4.2). The power is transmitted from the motor to a driving arm by means of a belt, pulley, and gear reduction box. A flexible wire guided by a pulley at each end of the platform, runs from the driving arm to each end of the polishing head. Tension on the wire is just sufficient to take up any slack. The wool cloth polishing head is 3⅜ inches long, 2 inches wide and one-half inch high. The polishing head is reciprocated through a distance of 13 inches in a straight path over the length of the test panel at approximately 37 oscillations (74 strokes) per minute.

4.4.4.1.4 *Polishing head.* The polishing head shall be the plated brass Abrasion Boat Attachment number 1703 for the Gardner Straight Line Washability and Abrasion Machine (see 6.4.3) of equivalent device consisting of two plated brass plates in which provisions have been made for alignment both between the upper and lower plates and with the flexible wire described in 4.4.4.1.3. The assembly shall weigh 1 ± 0.05 pounds.

4.4.4.1.5 *Wool cloth.* Cloth, 100 percent virgin wool, unbleached, approximately 47 ends and 41 picks per inch and weighing approximately 6 ounces per square yard. Botany wool style 404 or equal.

4.4.4.1.6 *Cleaning standard.* A freshly prepared mixture of equal weights of calcium carbonate conforming to JAN–C–293 and distilled water shall be used as the cleaning standard.

4.4.4.2 *Procedure.*

4.4.4.2.1 *Gloss determinations.* Specular gloss (45 degree) determinations shall be accomplished with the Photovolt Glossmeter Model 660 including the 45-degree 660M search unit, or equivalent. It is preferable to standardize the glossmeter using a high gloss working standard consisting of a mirror of reasonable stability. If such a mirror is not available, the Carrara black gloss standard furnished with the Photovolt 660M search unit can be used for standardizing the glossmeter in accordance with the Photovolt operating instructions. The gloss-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

meter standardization shall be checked with the standard before each gloss determination. All gloss readings shall be taken at the same position on the test panel.

4.4.4.2.2 *Polishing mirror.* Determine the gloss of the mirror according to 4.4.4.2.1. Place the mirror in the polishing apparatus in a position such that the length of the mirror is parallel to the motion of the polishing head. The head shall pass over the entire panel surface. Place metal or other material of suitable thickness at both ends of the test panel to hold it in place and provide a level surface for the polishing head to travel. Place two 2 by 6 inch layers of wool cloth in the abrasion boat. The surface of the fabric shall be smooth. Use a soft bristle brush to apply an even coating of the standard cleaner on the mirror. Polish for 50 cycles (100 strokes). Maintain an even coating of the cleaner on the mirror surface at all times. Reapplication may be required during the 50 cycles. After 50 cycles remove any polish remaining on the mirror with a lintless wiping tissue. Determine the gloss of the polished surface. Repeat this procedure until a constant value is achieved. With new wool cloth in the abrasion boat and the mirror polished with the standard clean, repeat the above procedure with the test polish. If after 250 cycles (500) strokes) the gloss reading has not increased by 16 units, a failure shall be recorded. Inspect the polished surface. Any residue remaining on the mirror or any scratches observed after cleaning with the metal polish shall constitute failure.

5. PREPARATION FOR DELIVERY

5.1 *Packaging.* Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* The metal polish shall be unit packaged in one-pint, screwcap bottles conforming to group A, class 1, style 1, closure A, and seal A of PPP–C–186.

5.1.1.2 *Intermediate packaging.* Twelve bottles of metal polish, unit packaged as specified in 5.1.1.1, shall then be packaged in accordance with the intermediate packaging requirements of PPP–C–186.

5.1.2 *Level C.* The metal polish shall be packaged in accordance with the industry's practice.

5.2 *Packing.* Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Forty-eight bottles of metal polish, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP–B–636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 *Level B.* Forty-eight bottles of metal polish, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP–B–636. All flaps of the fiber-board shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Metal polish, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 *Marking.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The metal polish covered by this specification is intended for cleaning searchlight metal mirrors but may be used for cleaning any metal mirrors. The polish is suitable for use at low environmental temperatures.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

c. Whether preproduction sample is required (see 3.1).

6.3 Suspending agent. A suspending agent may be added if the solids settle too rapidly to prevent the filling of containers with a homogeneous mixture.

6.3.1 The following suspending agents referenced in 3.4 are acceptable:

Cab-o-sil	Cabot Corporation, Boston, Massachusetts
Silene-D	Pittsburgh Plate Glass Company, Chemical Division, Pittsburgh, Pennsylvania
Hi-Sil 233	Pittsburgh Plate Glass Company, Chemical Division, Pittsburgh, Pennsylvania
Zeofree	Huber Corporation, New York, New York
Santocil Z	Monsanto Chemical Company, St. Louis, Missouri

Custodians:

Army — GL

Air Force — 69

Review activities:

Army — GL, MD

Air Force — 69

6.4 Testing material and equipment.

6.4.1 *Glossmeter.* Photovolt Photoelectric *Glossmeter Model 660 with the* Photovolt 660M Search Unit referenced in 4.4.4.1.2 can be obtained from Photovolt Corporation, New York 10, New York.

6.4.2 *Polishing apparatus.* Washability machine, referenced in 4.4.4.1.3 can be obtained from Gardner Laboratories, Inc., Bethesda, Maryland.

6.4.3 *Polishing head.* Abrasion Boat Attachment, referenced in 4.4.4.1.4 can be obtained from Gardner Laboratories, Inc., *Bethesda, Maryland.*

6.5 Notations. The outside margins of this document have been marked to indicate where changes from the previous issue have been made. An asterisk (*) indicates a substantive change in the paragraph text. A double asterisk (**) indicates deletion of a paragraph bearing the same number. The sign (#) indicates the addition of a completely new paragraph. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:

Army — GL

Project Number 7930–0057

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.

User activities:

Army — MU, MD, SM

Reviewer and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing Of DOD Documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:11Z
Check the source to verify that this is the current version before use.